

The bundles above imply a very interesting stability property. In the real case, the inclusion $O(n-1) \hookrightarrow O(n)$ induces an isomorphism on π_i for $i < n-2$, from the long exact sequence of the first bundle. Hence the groups $\pi_i O(n)$ are independent of n if n is sufficiently large, and the same is true for the groups $\pi_i U(n)$ and $\pi_i Sp(n)$ via the other two bundles. One of the most surprising results in all of algebraic topology is the Bott Periodicity Theorem which asserts that these stable groups repeat periodically, with a period of eight for O and Sp and a period of two for U . Their values are given in the following table:

$i \bmod 8$	0	1	2	3	4	5	6	7
$\pi_i O(n)$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	0	$\mathbb{Z}$	0	0	0	$\mathbb{Z}$
$\pi_i U(n)$	0	$\mathbb{Z}$	0	$\mathbb{Z}$	0	$\mathbb{Z}$	0	$\mathbb{Z}$
$\pi_i Sp(n)$	0	0	0	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	0	$\mathbb{Z}$

Stable Homotopy Groups

We showed in Corollary 4.24 that for an n -connected CW complex X , the suspension map $\pi_i(X) \rightarrow \pi_{i+1}(SX)$ is an isomorphism for $i < 2n+1$. In particular this holds for $i \leq n$ so SX is $(n+1)$ -connected. This implies that in the sequence of iterated suspensions

$$\pi_i(X) \rightarrow \pi_{i+1}(SX) \rightarrow \pi_{i+2}(S^2X) \rightarrow \cdots$$

all maps are eventually isomorphisms, even without any connectivity assumption on X itself. The resulting **stable homotopy group** is denoted $\pi_i^s(X)$.

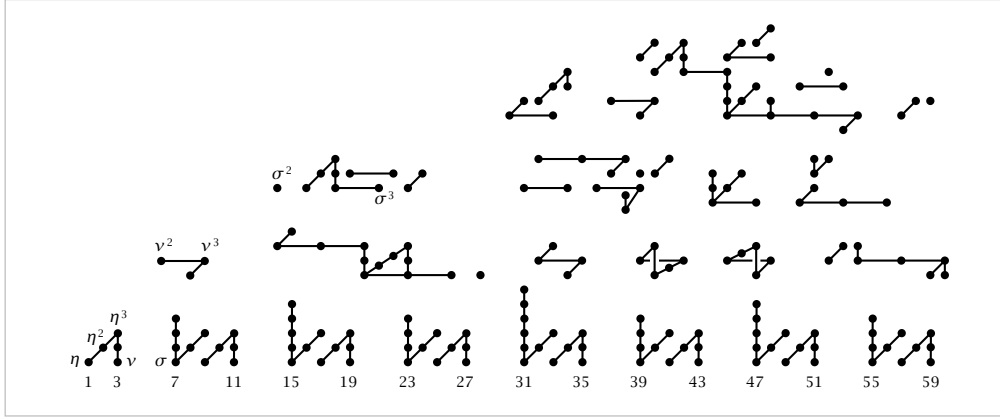
An especially interesting case is the group $\pi_i^s(S^0)$, which equals $\pi_{i+n}(S^n)$ for $n > i+1$. This stable homotopy group is often abbreviated to π_i^s and called the **stable i -stem**. It is a theorem of Serre which we prove in [SSAT] that π_i^s is always finite for $i > 0$.

These stable homotopy groups of spheres are among the most fundamental objects in topology, and much effort has gone into their calculation. At the present time, complete calculations are known only for i up to around 60 or so. Here is a table for $i \leq 19$, taken from [Toda 1962]:

i	0	1	2	3	4	5	6	7	8	9	10	11	12
π_i^s	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}_{24}$	0	0	$\mathbb{Z}_2$	$\mathbb{Z}_{240}$	$\mathbb{Z}_2 \times \mathbb{Z}_2$	$\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$	$\mathbb{Z}_6$	$\mathbb{Z}_{504}$	0
	13	14			15		16		17		18		19
	$\mathbb{Z}_3$	$\mathbb{Z}_2 \times \mathbb{Z}_2$			$\mathbb{Z}_{480} \times \mathbb{Z}_2$		$\mathbb{Z}_2 \times \mathbb{Z}_2$		$\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$		$\mathbb{Z}_8 \times \mathbb{Z}_2$		$\mathbb{Z}_{264} \times \mathbb{Z}_2$

Patterns in this apparent chaos begin to emerge only when one projects π_i^s onto its **p -components**, the quotient groups obtained by factoring out all elements of order relatively prime to the prime p . For $i > 0$ the p -component ${}_p\pi_i^s$ is of course isomorphic to the subgroup of π_i^s consisting of elements of order a power of p , but the quotient viewpoint is in some ways preferable.

The figure below is a schematic diagram of the 2-components of π_i^S for $i \leq 60$. A vertical chain of n dots in the i^{th} column represents a $\mathbb{Z}_{2^n}$ summand of π_i^S . The bottom dot of such a chain denotes a generator of this summand, and the vertical segments denote multiplication by 2, so the second dot up is twice a generator, the next dot is four times a generator, and so on. The three generators η , ν , and σ in dimensions 1, 3, and 7 are represented by the Hopf bundle maps $S^3 \rightarrow S^2$, $S^7 \rightarrow S^4$, $S^{15} \rightarrow S^8$ defined in Examples 4.45, 4.46, and 4.47.



The horizontal and diagonal lines in the diagram provide some information about compositions of maps between spheres. Namely, there are products $\pi_i^S \times \pi_j^S \rightarrow \pi_{i+j}^S$ defined by compositions $S^{i+j+k} \rightarrow S^{j+k} \rightarrow S^k$.

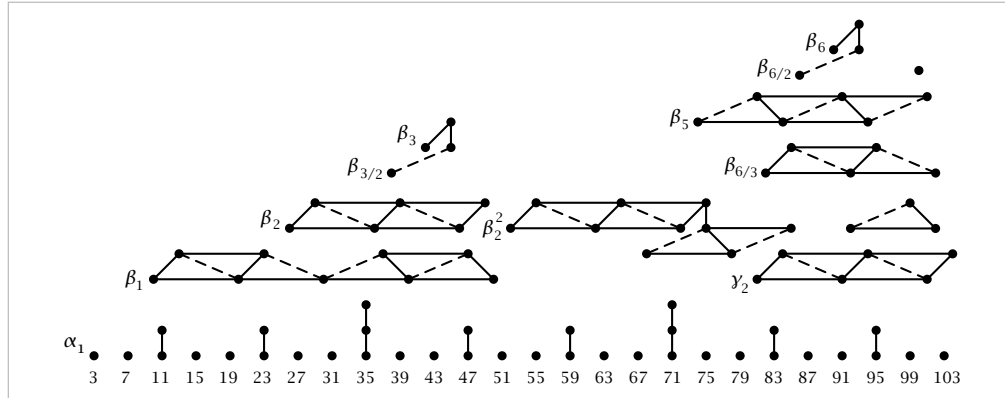
Proposition 4.56. *The composition products $\pi_i^S \times \pi_j^S \rightarrow \pi_{i+j}^S$ induce a graded ring structure on $\pi_*^S = \bigoplus_i \pi_i^S$ satisfying the commutativity relation $\alpha\beta = (-1)^{ij}\beta\alpha$ for $\alpha \in \pi_i^S$ and $\beta \in \pi_j^S$.*

This will be proved at the end of this subsection. It follows that ${}_p\pi_*^S$, the direct sum of the p -components ${}_p\pi_i^S$, is also a graded ring satisfying the same commutativity property. In ${}_2\pi_i^S$ many of the compositions with suspensions of the Hopf maps η and ν are nontrivial, and these nontrivial compositions are indicated in the diagram by segments extending 1 or 3 units to the right, diagonally for η and horizontally for ν . Thus for example we see the relation $\eta^3 = 4\nu$ in ${}_2\pi_3^S$. Remember that ${}_2\pi_3^S \approx \mathbb{Z}_8$ is a quotient of $\pi_3^S \approx \mathbb{Z}_{24}$, where the actual relation is $\eta^3 = 12\nu$ since $2\eta = 0$ implies $2\eta^3 = 0$, so η^3 is the unique element of order two in this $\mathbb{Z}_{24}$.

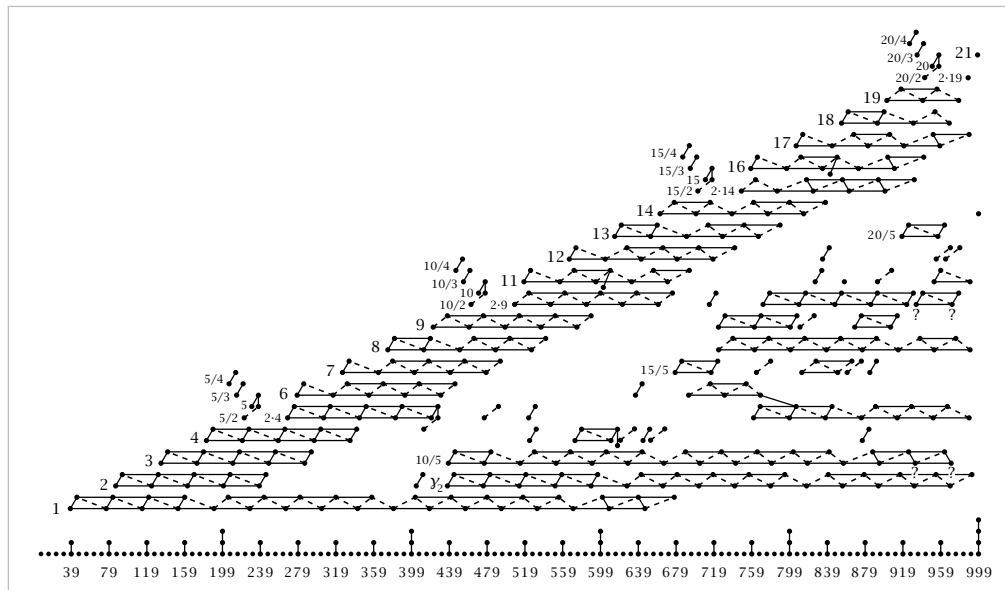
Across the bottom of the diagram there is a repeated pattern of pairs of ‘teeth.’ This pattern continues to infinity, though with the spikes in dimensions $8k - 1$ not all of the same height, namely, the spike in dimension $2^m(2n + 1) - 1$ has height $m + 1$. In the upper part of the diagram, however, there is considerably less regularity, and this complexity seems to persist in higher dimensions as well.

The next diagram shows the 3-components of π_i^S for $i \leq 100$, and the increase in regularity is quite noticeable. Here vertical segments denote multiplication by 3 and

the other solid segments denote composition with elements $\alpha_1 \in {}_3\pi_3^s$ and $\beta_1 \in {}_3\pi_{10}^s$. The meaning of the dashed lines will be explained below. The most regular part of the diagram is the ‘triadic ruler’ across the bottom. This continues in the same pattern forever, with spikes of height $m + 1$ in dimension $4k - 1$ for 3^m the highest power of 3 dividing $4k$. Looking back at the $p = 2$ diagram, one can see that the vertical segments of the ‘teeth’ form a ‘dyadic ruler.’



Even more regularity appears with larger primes, beginning with the case $p = 5$ shown in the next diagram. Again one has the infinite ruler, this time a ‘pentadic’ ruler, but there is also much regularity in the rest of the diagram. The four dots with question marks below them near the right edge of the diagram are hypothetical: The calculations in [Ravenel 1986] do not decide whether these potential elements of ${}_5\pi_i^s$ for $i = 932, 933, 970$, and 971 actually exist.



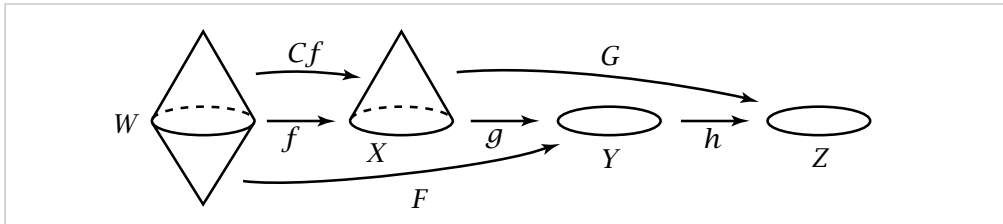
These three diagrams are drawn from tables published in [Kochman 1990] and [Kochman & Mahowald 1995] for $p = 2$ and [Ravenel 1986] for $p = 3, 5$.

For each p there is a similar infinite ‘ p -adic ruler,’ corresponding to cyclic subgroups of order p^{m+1} in ${}_p\pi_{2j(p-1)-1}^S$ for all j , where p^m is the highest power of p dividing j . These subgroups are the p -components of a certain cyclic subgroup of π_{4k-1}^S known as $\text{Im } J$, the image of a homomorphism $J: \pi_{4k-1}(O) \rightarrow \pi_{4k-1}^S$. There are also $\mathbb{Z}_2$ subgroups of π_i^S for $i = 8k, 8k + 1$ forming $\text{Im } J$ in these dimensions. In the diagram of ${}_2\pi_*^S$ these are the parts of the teeth connected to the spike in dimension $8k - 1$. The J -homomorphism will be studied in some detail in [VBKT].

There are a few other known infinite families in π_*^S , notably a family of elements $\beta_n \in {}_p\pi_{2(p^2-1)n-2p}^S$ for $p \geq 5$ and a family $\gamma_n \in {}_p\pi_{2(p^3-1)n-2p^2-2p+1}^S$ for $p \geq 7$. The element β_n appears in the diagram for $p = 5$ as the element in the upper part of the diagram labeled by the number n . These β_n ’s generate the strips along the upward diagonal, except when n is a multiple of 5 and the strip is generated by $\beta_2\beta_{n-1}$ rather than β_n . There are also elements β_n for certain fractional values of n . The element γ_2 generates the long strip starting in dimension 437, but $\gamma_3 = 0$. The element γ_4 in dimension 933 is one of the question marks. The theory behind these families β_n and γ_n and possible generalizations, as explained in [Ravenel 1986 & 1992], is one of the more esoteric branches of algebraic topology.

In π_*^S there are many compositions which are zero. One can get some idea of this from the diagrams above, where all sequences of edges break off after a short time. As a special instance of the vanishing of products, the commutativity formula in Proposition 4.56 implies that the square of an odd-dimensional element of odd order is zero. More generally, a theorem of Nishida says that every positive-dimensional element $\alpha \in \pi_*^S$ is nilpotent, with $\alpha^n = 0$ for some n . For example, for the element $\beta_1 \in {}_5\pi_{38}^S$ the smallest such n is 18.

The widespread vanishing of products in π_*^S can be seen as limiting their usefulness in describing the structure of π_*^S . But it can also be used to construct new elements of π_*^S . Suppose one has maps $W \xrightarrow{f} X \xrightarrow{g} Y \xrightarrow{h} Z$ such that the compositions gf and hg are both homotopic to constant maps. A nullhomotopy of gf gives an extension of gf to a map $F: CW \rightarrow Y$, and a nullhomotopy of hg gives an extension of hg to a map $G: CX \rightarrow Z$. Regarding the suspension SW as the union of two cones CW , define the **Toda bracket** $\langle f, g, h \rangle: SW \rightarrow Z$ to be the composition $G(Cf)$ on one cone and hF on the other.



The map $\langle f, g, h \rangle$ is not uniquely determined by f , g , and h since it depends on the choices of the nullhomotopies. In the case of π_*^S , the various choices of $\langle f, g, h \rangle$

range over a coset of a certain subgroup, described in an exercise at the end of the section.

There are also higher-order Toda brackets $\langle f_1, \dots, f_n \rangle$ defined in somewhat similar fashion. The dashed lines in the diagrams of ${}_3\pi_*^S$ and ${}_5\pi_*^S$ join an element x to a bracket element $\langle \alpha_1, \dots, \alpha_1, x \rangle$. Most of the unlabeled elements above the rulers in all three diagrams are obtained from the labeled elements by compositions and brackets. For example, in ${}_2\pi_*^S$ the 8-dimensional element is $\langle \nu, \eta, \nu \rangle$ and the 14-dimensional elements are σ^2 and $\langle \nu, \langle \nu, \eta, \nu \rangle, 2, \eta \rangle$.

Proof of 4.56: Only distributivity and commutativity need to be checked. One distributivity law is easy: Given $f, g: S^{i+j+k} \rightarrow S^{j+k}$ and $h: S^{j+k} \rightarrow S^k$, then $h(f + g) = hf + hg$ since both expressions equal hf and hg on the two hemispheres of S^{i+j+k} . The other distributivity law will follow from this one and the commutativity relation.

To prove the commutativity relation it will be convenient to express suspension in terms of smash product. The smash product $S^n \wedge S^1$ can be regarded as the quotient space of $S^n \times I$ with $S^n \times \partial I \cup \{x_0\} \times I$ collapsed to a point. This is the same as the quotient of the suspension S^{n+1} of S^n obtained by collapsing to a point the suspension of x_0 . Collapsing this arc in S^{n+1} to a point again yields S^{n+1} , so we obtain in this way a homeomorphism identifying $S^n \wedge S^1$ with S^{n+1} . Under this identification the suspension Sf of a basepoint-preserving map $f: S^n \rightarrow S^n$ becomes the smash product $f \wedge \mathbb{1}: S^n \wedge S^1 \rightarrow S^n \wedge S^1$. By iteration, the k -fold suspension $S^k f$ then corresponds to $f \wedge \mathbb{1}: S^n \wedge S^k \rightarrow S^n \wedge S^k$.

Now we verify the commutativity relation. Let $f: S^{i+k} \rightarrow S^k$ and $g: S^{j+k} \rightarrow S^k$ be given. We may assume k is even. Consider the commutative diagram below, where σ and τ transpose the two factors.

Thinking of S^{j+k} and S^k as smash products of circles, σ is the composition of $k(j+k)$ transpositions of adjacent circle

$$\begin{array}{ccccc} S^{i+k} \wedge S^{j+k} & \xrightarrow{f \wedge \mathbb{1}} & S^k \wedge S^{j+k} & \xrightarrow{\mathbb{1} \wedge g} & S^k \wedge S^k \\ & & \downarrow \sigma & & \downarrow \tau \\ & & S^{j+k} \wedge S^k & \xrightarrow{g \wedge \mathbb{1}} & S^k \wedge S^k \end{array}$$

factors. Such a transposition has degree -1 since it is realized as a reflection of the $S^2 = S^1 \wedge S^1$ involved. Hence σ has degree $(-1)^{k(j+k)}$, which is $+1$ since k is even. Thus σ is homotopic to the identity. Similarly, τ is homotopic to the identity. Hence $f \wedge g = (\mathbb{1} \wedge g)(f \wedge \mathbb{1})$ is homotopic to the composition $(g \wedge \mathbb{1})(f \wedge \mathbb{1})$, which is stably equivalent to the composition gf . Symmetrically, fg is stably homotopic to $g \wedge f$. So it suffices to show $f \wedge g \simeq (-1)^{ij} g \wedge f$. This we do

by the commutative diagram at the right, where σ and τ are again the transpositions of the two factors. As before, τ is homotopic to the identity, but now σ has

$$\begin{array}{ccc} S^{i+k} \wedge S^{j+k} & \xrightarrow{f \wedge g} & S^k \wedge S^k \\ \downarrow \sigma & & \downarrow \tau \\ S^{j+k} \wedge S^{i+k} & \xrightarrow{g \wedge f} & S^k \wedge S^k \end{array}$$

degree $(-1)^{(i+k)(j+k)}$, which equals $(-1)^{ij}$ since k is even. The composition $(g \wedge f)\sigma$ is homotopic to $(-1)^{ij}(g \wedge f)$ since additive inverses in homotopy groups are obtained by precomposing with a reflection, of degree -1 . Thus from the commutativity of the diagram we obtain the relation $f \wedge g \simeq (-1)^{ij} g \wedge f$. $\square$

Exercises

1. Use homotopy groups to show there is no retraction $\mathbb{R}P^n \rightarrow \mathbb{R}P^k$ if $n > k > 0$.
2. Show the action of $\pi_1(\mathbb{R}P^n)$ on $\pi_n(\mathbb{R}P^n) \approx \mathbb{Z}$ is trivial for n odd and nontrivial for n even.
3. Let X be obtained from a lens space of dimension $2n + 1$ by deleting a point. Compute $\pi_{2n}(X)$ as a module over $\mathbb{Z}[\pi_1(X)]$.
4. Let $X \subset \mathbb{R}^{n+1}$ be the union of the infinite sequence of spheres S_k^n of radius $1/k$ and center $(1/k, 0, \dots, 0)$. Show that $\pi_i(X) = 0$ for $i < n$ and construct a homomorphism from $\pi_n(X)$ onto $\prod_k \pi_n(S_k^n)$.
5. Let $f: S_\alpha^2 \vee S_\beta^2 \rightarrow S_\alpha^2 \vee S_\beta^2$ be the map which is the identity on the S_α^2 summand and which on the S_β^2 summand is the sum of the identity map and a homeomorphism $S_\beta^2 \rightarrow S_\alpha^2$. Let X be the mapping torus of f , the quotient space of $(S_\alpha^2 \vee S_\beta^2) \times I$ under the identifications $(x, 0) \sim (f(x), 1)$. The mapping torus of the restriction of f to S_α^2 forms a subspace $A = S^1 \times S_\alpha^2 \subset X$. Show that the maps $\pi_2(A) \rightarrow \pi_2(X) \rightarrow \pi_2(X, A)$ form a short exact sequence $0 \rightarrow \mathbb{Z} \rightarrow \mathbb{Z} \oplus \mathbb{Z} \rightarrow \mathbb{Z} \rightarrow 0$, and compute the action of $\pi_1(A)$ on these three groups. In particular, show the action of $\pi_1(A)$ is trivial on $\pi_2(A)$ and $\pi_2(X, A)$ but is nontrivial on $\pi_2(X)$.
6. Show that the relative form of the Hurewicz theorem in dimension n implies the absolute form in dimension $n - 1$ by considering the pair (CX, X) where CX is the cone on X .
7. Construct a CW complex X with prescribed homotopy groups $\pi_i(X)$ and prescribed actions of $\pi_1(X)$ on the $\pi_i(X)$'s.
8. Show the suspension of an acyclic CW complex is contractible.
9. Show that a map between simply-connected CW complexes is a homotopy equivalence if its mapping cone is contractible. Use the preceding exercise to give an example where this fails in the nonsimply-connected case.
10. Let the CW complex X be obtained from $S^1 \vee S^n$, $n \geq 2$, by attaching a cell e^{n+1} by a map representing the polynomial $p(t) \in \mathbb{Z}[t, t^{-1}] \approx \pi_n(S^1 \vee S^n)$, so $\pi_n(X) \approx \mathbb{Z}[t, t^{-1}]/(p(t))$. Show $\pi'_n(X)$ is cyclic and compute its order in terms of $p(t)$. Give examples showing that the group $\pi_n(X)$ can be finitely generated or not, independently of whether $\pi'_n(X)$ is finite or infinite.
11. Let X be a connected CW complex with 1-skeleton X^1 . Show that $\pi_2(X, X^1) \approx \pi_2(X) \times K$ where K is the kernel of $\pi_1(X^1) \rightarrow \pi_1(X)$, a free group. Show also that $\pi'_2(X, X^1) \approx \pi'_2(X) \times K'$ where K' is the quotient of K obtained by factoring out the conjugation action of $\pi_1(X^1)$. Note that K' is abelian.
12. Show that a map $f: X \rightarrow Y$ of connected CW complexes is a homotopy equivalence if it induces an isomorphism on π_1 and if a lift $\tilde{f}: \tilde{X} \rightarrow \tilde{Y}$ to the universal covers induces an isomorphism on homology. [The latter condition can be restated in terms of